THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 5: **Describing Motion Around Us** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A car starting from rest has acceleration 3 m/s squared. Its velocity after 4 s is (m/s): 1

(a) 7 m/s **(b)** 1 m/s **(c)** 12 m/s **(d)** 6 m/s

2. A body moves along a straight line with speed 4 m/s for 10 s and 6 m/s for the next 10 s. The average speed is: 1

(a) 6 m/s **(b)** 10 m/s **(c)** 5 m/s **(d)** 4 m/s

3. A stone thrown vertically up returns to the thrower's hand after 6 s. The velocity of throw is (g = 10): 1

(a) 10 m/s **(b)** 15 m/s **(c)** 60 m/s **(d)** 30 m/s

4. A body starts from rest with acceleration 2 m/s squared. The distance covered in 10 s is: 1

(a) 200 m **(b)** 100 m **(c)** 20 m **(d)** 10 m

5. A car travels half the distance at 40 km/h and the other half at 60 km/h. The average speed is: 1

(a) 52 km/h **(b)** 50 km/h **(c)** 45 km/h **(d)** 48 km/h

6. A body in circular motion of radius 2 m makes 1 revolution in 4 s. Speed is (pi = 3.14): 1

(a) 1.57 m/s **(b)** 3.14 m/s **(c)** 12.56 m/s **(d)** 6.28 m/s

7. A car brakes from 20 m/s to rest with retardation 5 m/s squared. The distance covered is: 1

(a) 40 m **(b)** 100 m **(c)** 10 m **(d)** 4 m

8. A graph of velocity against time is a straight line passing through the origin. The body has: 1

(a) uniform velocity **(b)** zero acceleration

(c) retardation **(d)** uniform acceleration from rest

9. The displacement of a body after half a circle of radius r is: 1

(a) r **(b)** zero **(c)** pi r **(d)** 2r

10. A car moving at 72 km/h stops in 8 s. The retardation is: 1

(a) 2.5 m/s squared **(b)** 0.9 m/s squared **(c)** 9 m/s squared **(d)** 4 m/s squared

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A car moves with uniform velocity of 15 m/s for 20 s. Find the distance and draw the nature of the v-t graph. 2

12. An object is dropped from 20 m. Find the time to hit the ground and the speed on impact (g = 10). 2

13. Why is the free fall of a feather and a stone in a vacuum identical? 2

14. A scooter changes its velocity from 6 m/s to 12 m/s in 3 s. Find the acceleration and the distance moved. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A bus moving at 20 m/s speeds up uniformly to 30 m/s in 5 s. Find the acceleration, the distance and the average velocity. 3

16. A car A starts from rest with a = 2 m/s squared and car B moves at uniform 20 m/s. When will A overtake B if both start together from the same point? 3

17. An object has a uniform velocity of 8 m/s for 5 s, rests for 5 s and then moves with 4 m/s for 10 s. Find the total distance and the average speed. 3

OR

A ball is thrown upward with 25 m/s. Find the time to rise, the maximum height, and the velocity at 1 s (g = 10). 3

18. The speed-time data of a car are: t(s) 0, 2, 4, 6 and v(m/s) 0, 6, 12, 18. Find the acceleration and the distance in 6 s. 3

OR

Show that for uniform acceleration the average velocity equals (u + v)/2 and use it to derive s = (u + v)t/2. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A car accelerates from rest at 2 m/s squared for 10 s, moves at constant velocity for 20 s and then retards uniformly to rest in 5 s. Find the maximum velocity, the retardation and the total distance. Draw the v-t graph in words. 5

OR

A driver at 72 km/h sees an obstacle 50 m ahead and brakes with a retardation of 5 m/s squared after a reaction time of 0.5 s. Will he stop in time? 5

20. A ball is thrown vertically upward from the ground with 30 m/s. Find 5
(a) the time to reach the top
(b) the maximum height
(c) the total time of flight
(d) the velocity on returning (g = 10).

OR

Explain uniform circular motion of the Moon around the Earth: the orbital period 27 days, orbital radius 3.84 x 10^5 km, find the orbital speed in km/s (pi = 3.14). 5

END OF QUESTION PAPER — Total: 40 Marks